

District 7 SPSS Comparison 2023 - 2025 Sales

Elliot Dean - Program Analyst IV
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

4 May 2026

Contents

Model I	1
Model II	1
Model III	2
Model IV	2
Model V	2
Model VI	3

Model I

Table 1: District 7 | LN_SPPSF Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1385	1372	13	0.4507	0.1380	0.1300	16.9220
R	1385	1372	13	0.4529	0.1298	0.1216	15.7416
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.48%	-5.94%	-6.46%	-6.98%

Model II

Table 2: District 7 | LN_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1805	1790	15	0.4652	0.2250	0.2190	34.6630
R	1805	1790	15	0.4680	0.2159	0.2094	32.8677
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.59%	-4.04%	-4.38%	-5.18%

Model III

Table 3: District 7 | LN_PRICE Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1805	1779	26	0.4302	0.3420	0.3320	35.4870
R	1805	1779	26	0.4317	0.3368	0.3272	34.7546
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.35%	-1.52%	-1.45%	-2.06%

Table 4: District 7 | LN_PRICE Model Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	0.9890	1.0970	0.9140	1.2000	0.3760	
R	0.9855	1.0974	0.9137	1.2011	0.3771	-0.4988
% diff (R vs SPSS)	-0.35%	0.04%	-0.03%	0.09%	0.28%	

Model IV

Table 5: District 7 | LN_PRICE (Inliers) Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1623	1598	25	0.3668	0.4260	0.4170	47.4560
R	1647	1621	26	0.3748	0.3936	0.3839	40.4716
% diff (R vs SPSS)	1.48%	1.44%	4.00%	2.18%	-7.61%	-7.94%	-14.72%

Table 6: District 7 | LN_PRICE (Inliers) Model Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	1.0070	1.0680	0.9380	1.1390	0.3130	-0.3640
R	0.9961	1.0717	0.9358	1.1452	0.3230	-0.4079
% diff (R vs SPSS)	-1.08%	0.35%	-0.23%	0.55%	3.19%	-12.05%

Model V

Table 7: District 7 | LN_TASP Model I Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1623	1599	24	0.3667	0.4020	0.3930	44.7940
R	1647	1622	25	0.3751	0.3676	0.3579	37.7134
% diff (R vs SPSS)	1.48%	1.44%	4.17%	2.30%	-8.56%	-8.93%	-15.81%

Table 8: District 7 | LN_TASP Model I Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	1.0070	1.0680	0.9370	1.1400	0.3120	-0.3900
R	1.0606	1.1497	0.9954	1.1550	0.3326	-0.4647
% diff (R vs SPSS)	5.33%	7.65%	6.23%	1.31%	6.61%	-19.15%

Model VI

Table 9: District 7 | LN_TASP Model II Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	2483	2452	31	0.4038	0.3920	0.3850	51.0820
R	1647	1625	22	0.3761	0.3629	0.3543	42.0724
% diff (R vs SPSS)	-33.67%	-33.73%	-29.03%	-6.86%	-7.42%	-7.97%	-17.64%

Table 10: District 7 | LN_TASP Model II Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	0.9970	1.0840	0.9280	1.1680	0.3480	-0.4410
R	1.0600	1.1502	0.9948	1.1563	0.3337	-0.4713
% diff (R vs SPSS)	6.32%	6.11%	7.19%	-1.00%	-4.12%	-6.87%